



Introduction to Blockchain

CSE-355

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Lecture 01

Tuesday, Jan 20, 2026

Bitcoin Statistics: Jan 22, 2026

- Bitcoins in circulation: 19.98 Million (over \$2T market cap)
- Total will be 21 Million (last will be mined about in the year 2140)
- Bitcoins left to be mined: 1.02 Million (Total issued: 95.1%)
- Mined Bitcoin Blocks: 933,250 yesterday; 933,390 (this morning)
- Blockchain Size: 715 GB
- New Bitcoin Blocks per day (presently): 144 (on average)
- New Bitcoins per day (presently): 450 (144×3.125)
- About 4 Million Bitcoins were lost forever (estimate)
- Transactions per second: 7 (VISA has about 24k)
- Yet, Bitcoin is only one of the many cryptocurrencies around

The Cryptocurrency Avalanche

- Cryptocurrencies today: 18000+ active (lot more inactive)
 - 250+ on Coinbase and 400+ on Binance
- Cryptocurrency users: (500-600)+ Million
- Businesses accepting crypto payments: ~250+ listed + Paypal access
- Crypto exchanges: 1000+ (some failed, some big failures!!!)
- Despite the humongous growth in cryptocurrencies:
 - We will not cover crypto for the reasons they are famous for
- Our focus: Blockchain Technology (concepts around for 30 years)
- Or, more broadly, distributed ledger technology (DLT)
 - Around for about 40 years

Blockchain does not always mean Bitcoin

What is a Blockchain?

- 1) A practical introduction to the basics of blockchain
- 2) Get familiar with the common terminology and mechanisms

About this course

- What will this course teach you?
 - Underlying Technologies to Construct a Blockchain
 - Blockchains: Properties and Pitfalls
 - Programmability in Blockchains: Smart Contracts
 - Cryptocurrencies – as a Use Case for Blockchain
 - Financial Applications: Stablecoins and Asset Tokenization
- What it won't?
 - Steps to open a crypto account in a given regime
 - Investment strategies [on your own or with an exchange]
 - What kind of regulations should be adopted

Why should one take this course?

Motivation for creating Blockchain

The growing need for digital payments led to challenges

- Requiring intermediaries

- Which can prohibit or reverse payment
- May not always be online

Needed direct
P2P payments

- The matter of double-spending

- **Discussion:** Can you double-spend physical currency notes?
- Inherent problem in payments that are both: **P2P** **AND** **digital**

- Missing financial freedom (the bailouts of 2008)

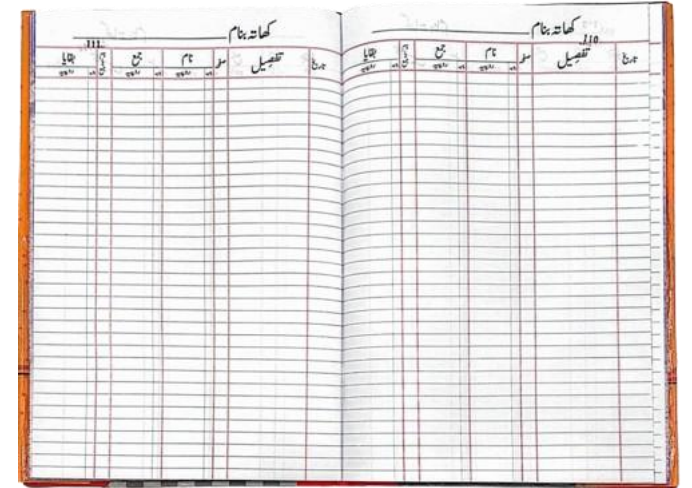
- Complicated global transactions (we need: fast and cheap)

Satoshi Nakamoto created blockchain technology in 2009

What is Blockchain?

- A ledger for recording data, just like databases
 - Used to record **transactions** (often financial!)
 - Across a network of computers (**nodes**)

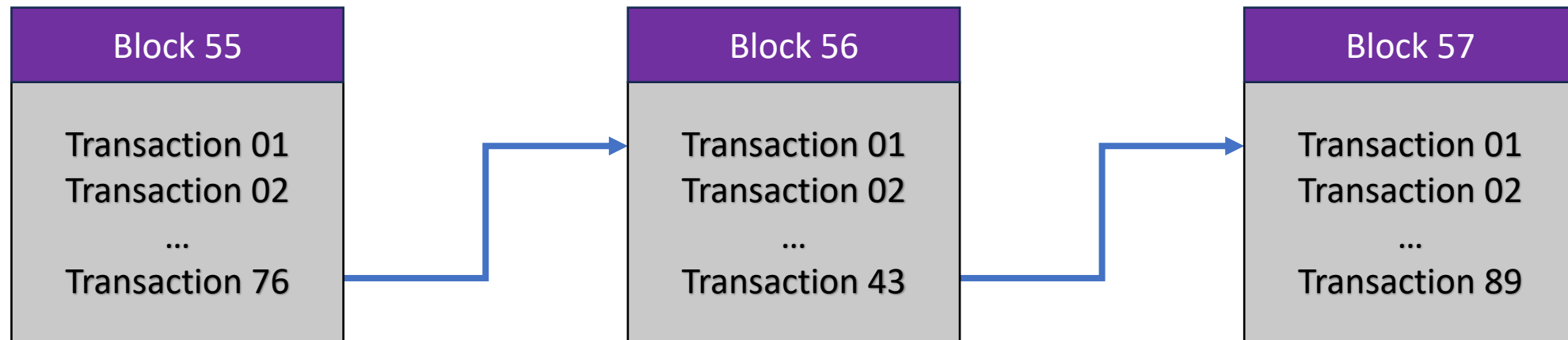
The distributed ledger



- Who initiates/generates those transactions?
 - Any node (*a computer running software*) in the blockchain network
 - An end user or an organization (e.g., an exchange) on their behalf
- Who maintains/stores the distributed ledger?
 - Most nodes keep a part of the ledger
 - Some nodes keep a full ledger

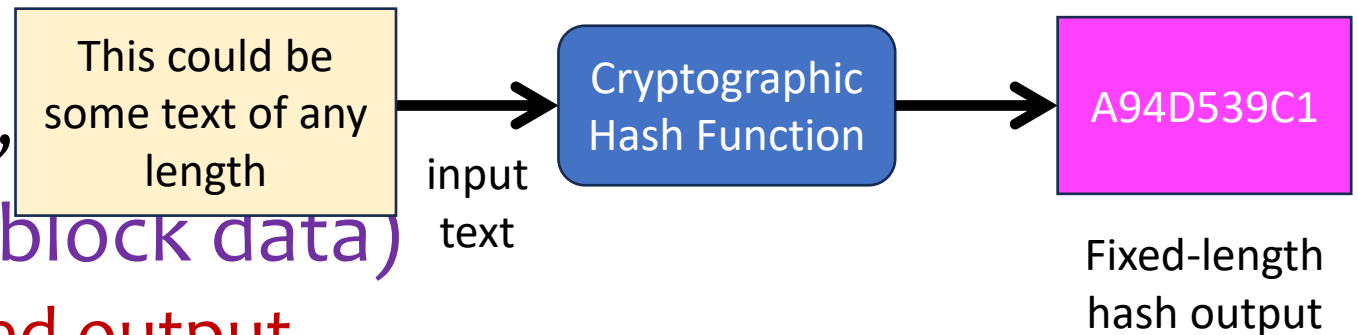
Why is it called a Blockchain?

- Because of its structure
 - Transactions are written in **blocks** of data
 - Which are somehow **chained** together
 - Through cryptographic hashing
 - And what is that?



Cryptographic Hashing

- Maps an input text to a fixed-length, random-looking output
 - A complicated mathematical function – almost black magic
- How is *hash* computed?
 - Well-known software (e.g.,
- Any change in input text (block data)
 - Significant change in hashed output

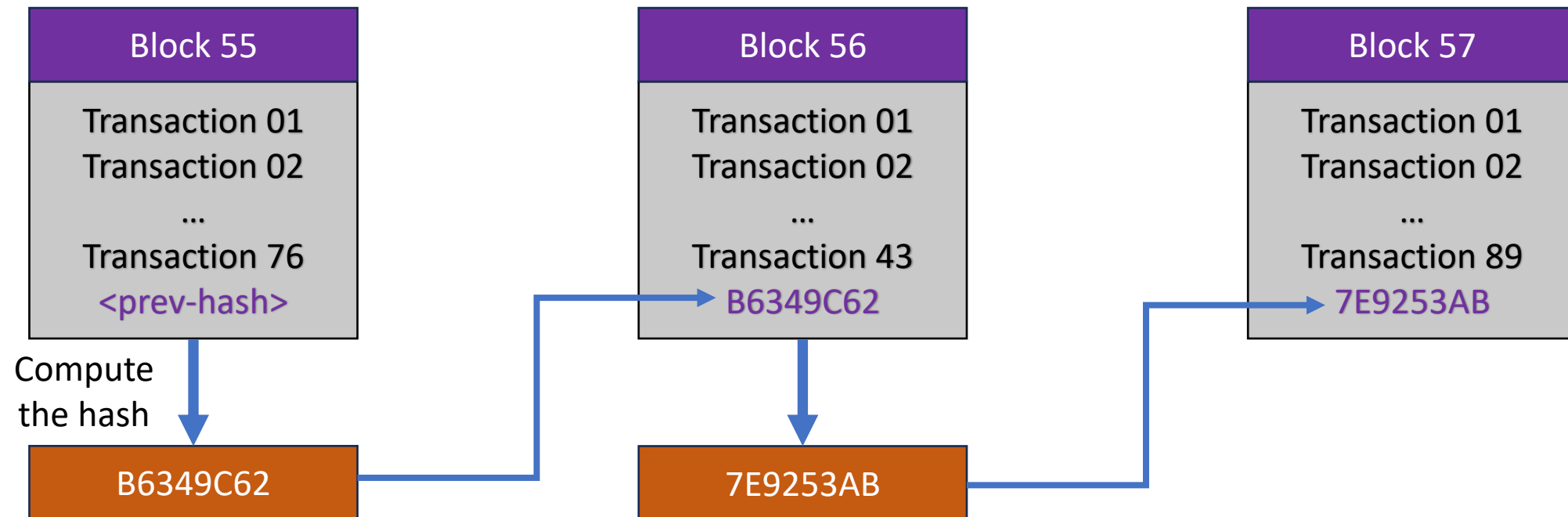


Input your text here:

SHA256 hash: be98c2510e417405647facb89399582fc499c3de4452b3014857f92e6baad9a9

Chaining Blocks using Hashes

- Each block contains a **hash (or digest)** of the previous block



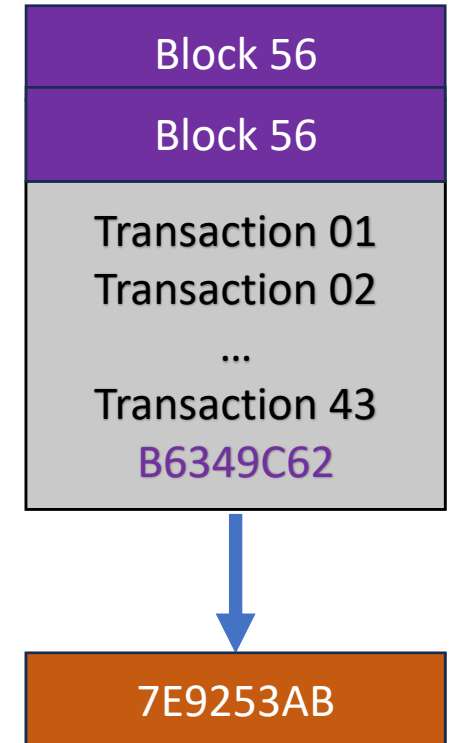
Hashes in a Practical Blockchain

- Hashes need to meet **some** criteria
 - For example, a hash must start with 3 zeros
 - What if it does not meet the criteria?
 - The block will not be allowed
- What to do?
- Change something in the block data
 - But block data is fixed (Transactions)
 - Include a fictitious number (NONCE)
 - What if the hash value **still** does not meet the criteria?

Keep changing NONCE until criteria met



MINING!

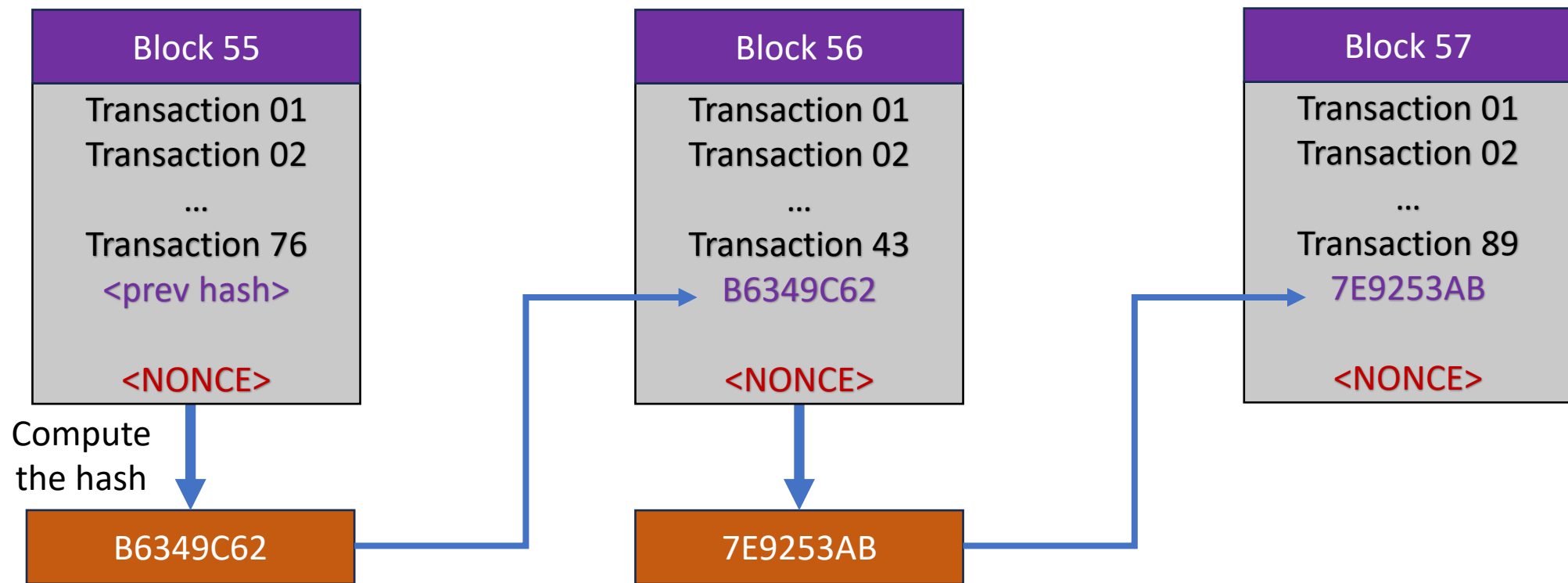


Lecture 02

Thursday, Jan 22, 2025

Recap: Transactions, Blocks, Hashes

- Each block in a typical blockchain contains:
1) Transactions, 2) hash of previous block, and 3) NONCE



Mining: Cycle through the NONCE to meet *some* hash criteria

What does Blockchain bring to the table?

- Immutability / tamper-proof
 - Nearly impossible to change or roll back transactions
 - Ensured by hashes – we make it very hard to (re)compute the hash
 - Change a transaction: recompute hash (this block **AND** all subsequent ones)

What if someone actually does that?

- Auditability
 - All transactions can be tracked and verified
 - All historical transactions are around and available

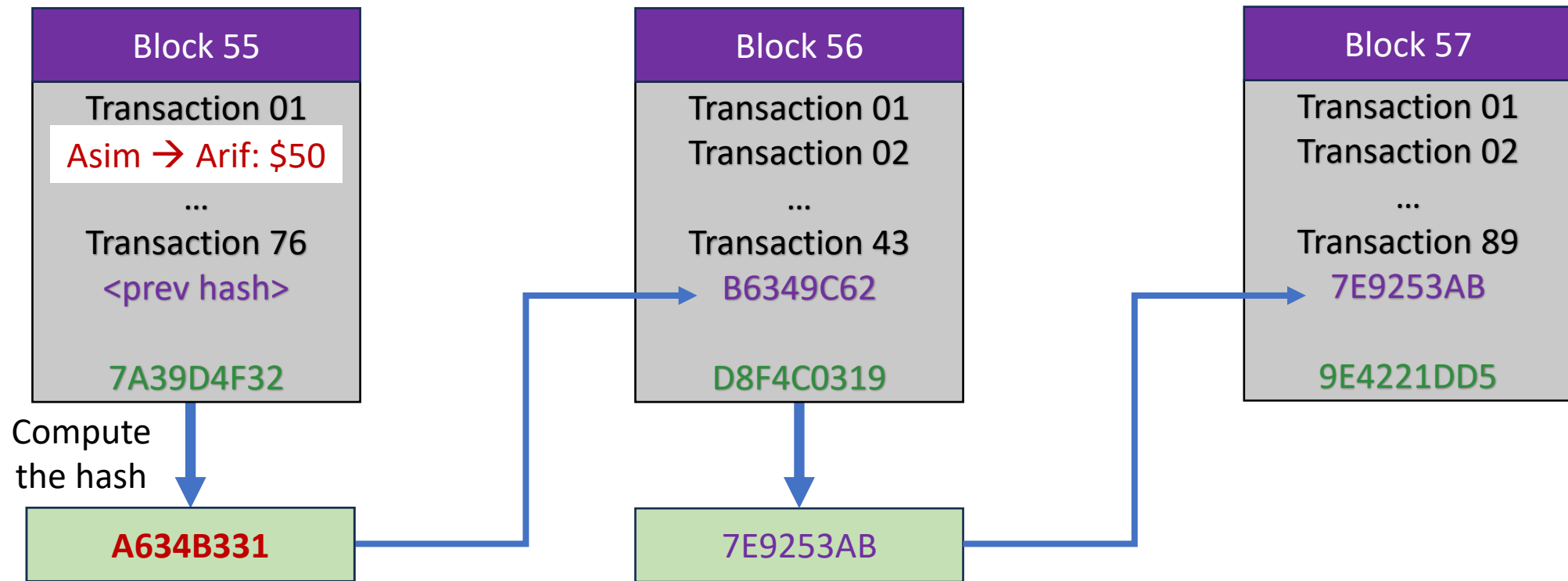
Privacy?

Anonymity?

- Decentralization
 - Enables autonomy and censorship resistance
 - Transactions without needing a central authority (such as banks!)
 - Reduced risk of single points of failure

Securing + Chaining of Blocks

- What if the data in a block is **tampered** with?
- Hash value changes: leads to **Unchaining**



Any change in the block data can be easily detected

Security and Efficiency in blockchain

- Can the data written on blockchain be **permanently** tampered?
 - Possible, but requires more than 50% of control
- Security of **financial transactions**:
 - Has the payer authorized the transaction?
 - Is the payee the right person?
- Can we verify transactions before writing them in a block?
 - Is someone trying to spend more than what they have?
 - How do we check this in current banking?
- Can this verification be done efficiently?
 - Simplified Payment Verification

Digital Signatures and
Public Key Cryptography

Transactions
are all there

When to use a Blockchain?

- When there is a multitude of transactions that:
 - Are generated (and potentially consumed) by multiple parties
 - Have a common interest
 - Need to avoid a central control (**decentralization**)
 - **Discussion:** How can you do financial transactions without a bank?
 - Consider a tribe on an island a couple of thousand years ago!
- Boils down to a Multi-party **consensus** system
 - Everyone agrees that Muneeb has transferred PKR 70k to Zain
 - Or that food at a restaurant now ranks 3.1 out of 5
 - **Agreement over a shared set of data**

Consensus Options

- Recap: Why is consensus needed?
 - To develop a ledger (single converged view of blockchain)
- First popular consensus option: Proof of work (PoW)
 - Block creators purchase (and spend) energy to create blocks
 - In exchange, they get some **reward**
 - ***Block creators are called miners***
- Second popular consensus option: Proof of stake (PoS)
 - Block creators purchase (and stake) cryptocurrency
 - In exchange, they get a **fee**
 - ***Block creators are called validators***

Bitcoin
Blockchain

Ethereum
Blockchain

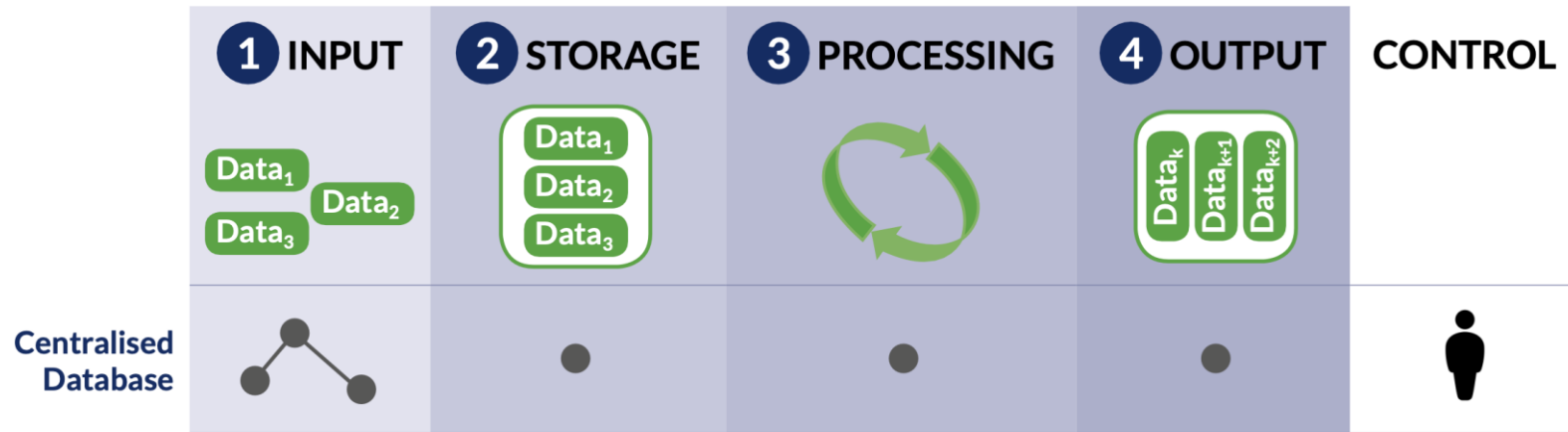
Validity of a blockchain

- When is a blockchain legit?
 - When all the blocks are valid
- How do we ensure that a block is valid?
 - A correct hash of the previous block is present
 - Hash of this block meets the criteria (PoW blockchain)
 - All transactions in the block are valid
- How do we ensure a transaction is valid?
 - Cryptocurrency example: Alice gives 0.3 BTC to Bob
 - Did Alice have the currency?
 - Did Alice agree?
 - How do we indicate that Alice agreed?

We will answer these questions soon!

Process: Blockchain Formation

Centralized Database vs Distributed Ledger

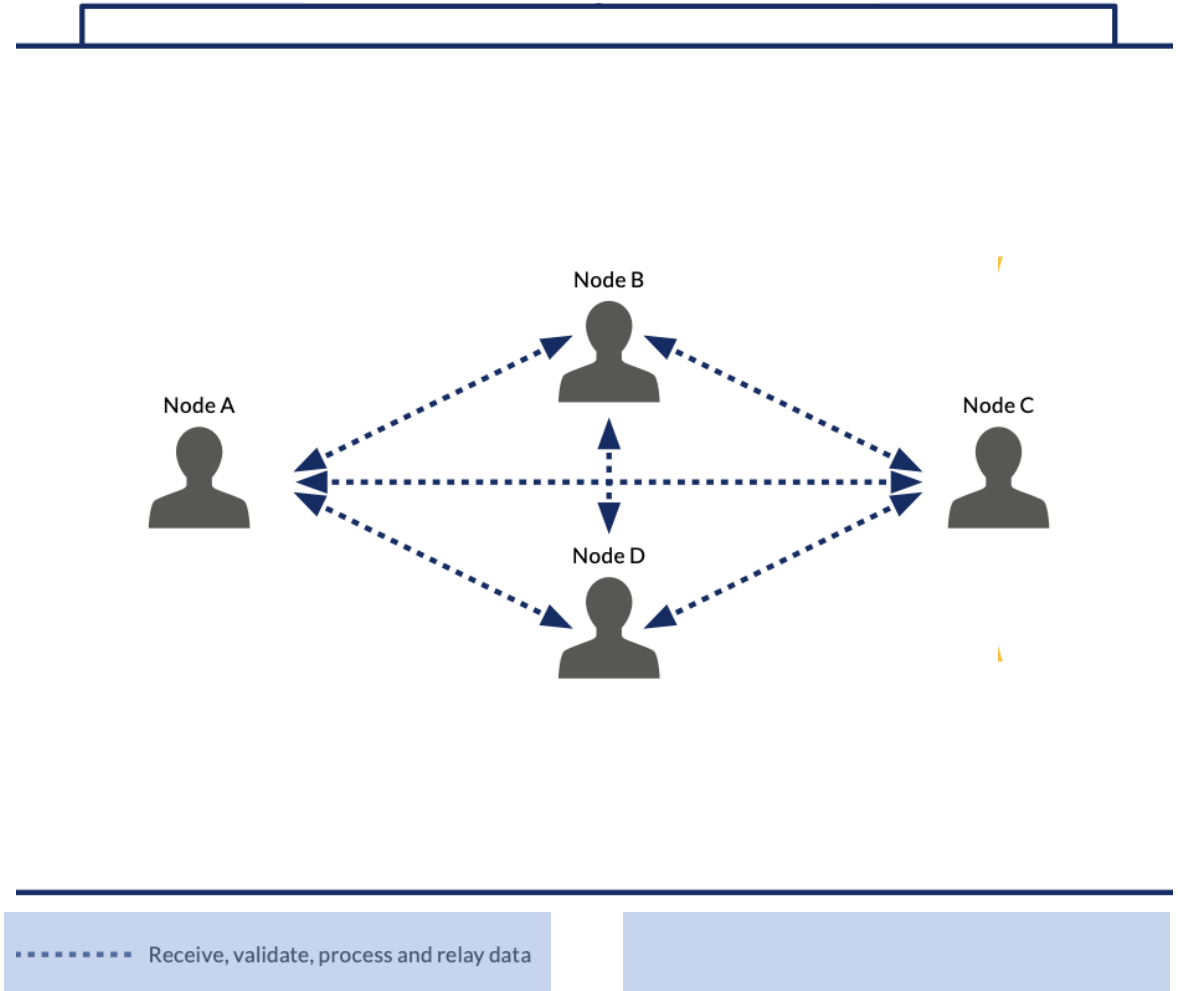


Legend: Single entity control Multiple entity control Single node Multiple nodes

DLT: An Accurate Description

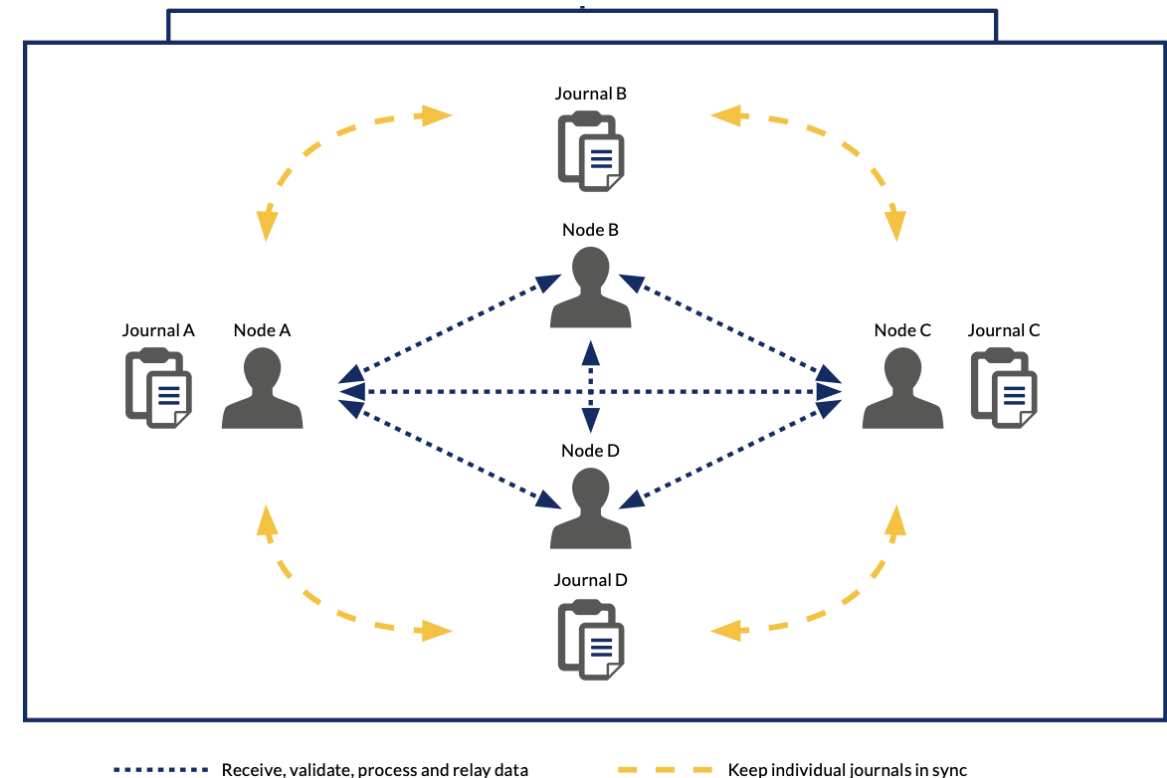
- Consider four entities/nodes distributed in the network
- Transaction data is **generated** by these nodes (**and shared**)
- Each node builds a **journal**
 - its own version of the database
 - For this example: 4 journals
 - Journals could be *different*; why?

The blockchain is the unified view of individual journals



DLT: An Accurate Description

- **GOAL:** Different journals are in sync *most* of the time
 - Offer a **shared view**
- Sync effort must be without a central party
- What is the **LEDGER**?
 - Not the database
 - Mechanism of syncing different journals



Syncing: Building consensus

- Set of journals
 - Individual views of shared data
- Convergence to a single shared view
 - That everyone agrees with – The distributed ledger
- A ledger isn't quite a tangible thing
 - Just emerges from the efforts of synchronizing individual journals
- Distributed consensus (Syncing): How does that work?
 - Five-step process, but before that:
 - Building distributed consensus is HARD!